La Scuola d'Italia / Dr. Huson / IB Math 11 (a)
9 November 2026

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2.10 Classwork: Systems of Equations and Exponents (no calculator)

Start in class and finish for homework. Write each step of your working on its own line.

1. Solve each system of equations by elimination.

   (a) $4x + 3y = 5 \quad (1)$
   $6x - 5y = 17 \quad (2)$

   (i) Multiply equation (1) by 3 and equation (2) by 2, so that both equations have $12x$. Write the two new equations. Multiply *every* term.

   (ii) Subtract one new equation from the other to eliminate x. Solve for y.

   (iii) Substitute your value of y into equation (1) to find x.

   (b) $5x + 2y = 3$
   $3x + 7y = 25$

2. **Finding two unknowns.** Each fact you are given becomes one equation. Two facts give two equations, and you solve the system.

In an arithmetic sequence, $u_4 = 17$ and $u_{10} = 41$.

(a) Use $u_n = u_1 + (n-1)d$ to write two equations in u_1 and d.

(b) Solve your equations to find u_1 and d.

(c) Find the sum of the first 20 terms.

The sum of the first n terms of a different arithmetic sequence is $S_n = pn^2 - qn$. You are given that $S_2 = 2$ and $S_5 = 35$.

(d) Substitute $n = 2$ to show that $4p - 2q = 2$.

(e) Use $S_5 = 35$ to write a second equation in p and q.

(f) Solve your two equations to find p and q.

(g) S_1 is the sum of just the first term, so $u_1 = S_1$. Find u_1.

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3. The curve $y = A\,(2^x) + B$ passes through the points $(1,\ 7)$ and $(3,\ 19)$.

   (a) Substitute $x = 1$, $y = 7$ to write an equation in A and B. Do the same for the point $(3,\ 19)$.

   (b) Solve the two equations to find A and B.

   (c) Find the value of y when $x = 5$.

   (d) Find the value of x when $y = 35$.

4. Find the integers a and b such that $\left(3 + \sqrt{2}\right)\left(a + b\sqrt{2}\right) = 4 - \sqrt{2}$.

   (a) Expand the left side. Collect it into a rational part and a $\sqrt{2}$ part.

   (b) The rational parts on the two sides are equal, and the $\sqrt{2}$ parts are equal. Write two equations in a and b, and solve them.

5. (a) Write each number as a power of 2, 3 or 5.

16	32	64	27	81	25	125	$\frac{1}{8}$	$\frac{1}{9}$	$\sqrt{5}$

(b) Simplify. A power of a power multiplies the exponents: $(a^m)^n = a^{mn}$.

(i) $(a^3)^2$ (ii) $(2x^4)^3$ (iii) $(3a^2b)^2 \times 2a^{-3}b$

(c) Solve each equation by writing both sides as powers of the same base.

(i) $2^{x+1} = 32$ (ii) $9^x = 27^{x-1}$ (iii) $5^{2x} = \dfrac{1}{125}$

(d) Consider the equation $3^{2x} - 10\,(3^x) + 9 = 0$.

(i) Let $u = 3^x$. Explain why $3^{2x} = u^2$, and write the equation in terms of u.

(ii) Solve the equation for u.

(iii) Hence find the two values of x.

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Start in class and finish for homework. Write each step of your working on its own line.

1. Solve each system of equations by elimination.

   (a) $5x + 2y = 3$
   $3x + 7y = 25$

   (b) $3x + 2y - z = 1 \quad (1)$
   $2x - 3y + 4z = 19 \quad (2)$
   $4x + 5y + 2z = 9 \quad (3)$

   Number every new equation. Eliminate the *same* variable twice, from two different pairs, to get equations (4) and (5) in two variables. Solve (4) and (5), then substitute back to find the third variable.

2. **Finding two unknowns.** Each fact you are given becomes one equation. Two facts give two equations, and you solve the system.

In an arithmetic sequence, $u_3 = 14$ and $S_8 = 148$.

(a) Write two equations in u_1 and d.

(b) Find u_1 and d.

(c) Find u_{20}.

The sum of the first n terms of a different arithmetic sequence is $S_n = pn^2 - qn$. You are given that $S_2 = 2$ and $S_5 = 35$.

(d) Find the value of p and the value of q.

(e) Find u_1 and u_2, and hence the common difference.

(f) Show that $u_n = 4n - 5$.

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3. The curve $y = A\left(2^x\right) + B$ passes through the points $(-1,\ 4)$ and $(2,\ 11)$.

   (a) Find the value of A and the value of B.

   (b) Show that the equation of the curve can be written as $y = 2^{x+1} + 3$.

   (c) Find the value of x when $y = 131$.

4. (a) Find the integers a and b such that $\left(2 + \sqrt{3}\right)\left(a + b\sqrt{3}\right) = 5 + 4\sqrt{3}$.

   (b) Hence write $\dfrac{5 + 4\sqrt{3}}{2 + \sqrt{3}}$ in the form $p + q\sqrt{3}$, where $p, q \in \mathbb{Z}$.

5. (a) Write each number as a power of 2, 3 or 5.

32	64	81	125	$\frac{1}{8}$	$\frac{1}{9}$	$\sqrt{5}$	$4\sqrt{2}$	$\frac{1}{5\sqrt{5}}$	$\sqrt[3]{9}$

(b) Simplify. Give your answers using positive exponents only.

(i) $(2x^4)^3$ (ii) $(3a^2b)^2 \times 2a^{-3}b^{-4}$ (iii) $\dfrac{(4a^6)^{\frac{3}{2}}}{(2a)^3}$

(c) Solve each equation.

(i) $25^{x+1} = 125^{x-1}$ (ii) $4^x \times 8^{x-1} = 32$ (iii) $27^x = \dfrac{1}{9\sqrt{3}}$

(d) By letting $u = 3^x$, solve the equation $9^x - 4\,(3^{x+1}) + 27 = 0$.